

★ 0/1 Knapsack using Least cost B&B

→ 0/1 Knapsack is Maximization problem
→ We can find Upper bound at the Node

→ Upperbound at Particular Node

=
Total profit at that particular Node
+

$(\text{Capacity} - \text{Total weight at that Particular Node}) * \frac{P}{W}$ of the Next Item

i.e. Next Highest Value

→ Here, Total profit at that Node describes Value obtained after deciding about objects up to that level, say i .

→ Capacity - Total weight at that Particular Node

the remaining Capacity.



By multiplying it with the next highest P/W , we get prediction on the Maximum value of profit that can be obtained (i.e. Upper bound)

∴ 2nd term gives Upperbound on the Profit

∴ We need to take Input such that items are sorted according to P/W in the decreasing order.

Ex 1

P	10	10	12	18
W	2	4	6	9
P/W	5	2.5	2	2

Capacity $m = 15$

→ $\textcircled{1} \begin{cases} CP = 0, CW = 0 \\ UB = 75 \end{cases}$

CP = Current Profit
CW = Current Weight

$$\begin{aligned} UB(1) &= \text{Current Profit} + (\text{Capacity} - \text{Current Weight}) \times \frac{P}{W} \\ &= 0 + (15 - 0) \times 5 \\ &= 75 \text{ i.e. Upper Bound on Profit is } 75 \end{aligned}$$

→ Set global variable Answer = $-\infty$
(∵ It is Maximization problem)

→ Once we get some better solution Answer can be improved by that greater Value.

→ Now, we have 2 options

Take 1st Item
 $X_1 = 1$

Don't take 1st Item
 $X_1 = 0$

→ Taking 1st Item
Makes $CP = 10$
 $CW = 2$

(Next i.e. P_2/w_2)

$$\begin{aligned} UB &= CP + (15 - CW) * \frac{P_2}{w_2} \\ &= 10 + (15 - 2) * \frac{10}{4} \\ &= 10 + \frac{13 * 5}{2} = 10 + 32.5 \\ &= \boxed{42.5} \end{aligned}$$

→ Not Taking 1st Item

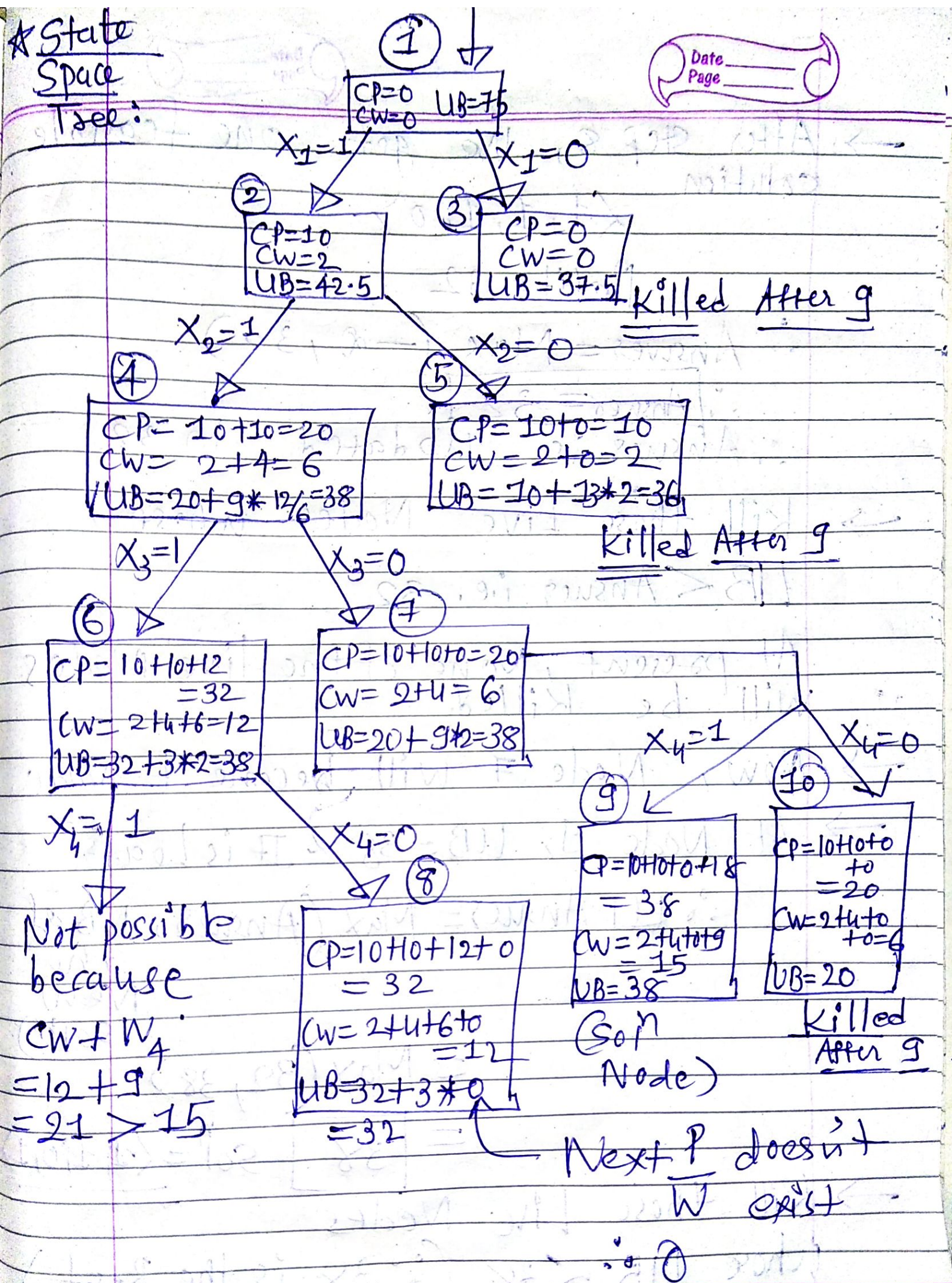
Makes $CP = 0$
 $CW = 0$

& Next P/w

$$\begin{aligned} UB &= CP + (15 - CW) * \frac{P_2}{w_2} \\ &= 0 + (15 - 0) * \frac{10}{4} \\ &= \frac{75}{2} \\ &= \boxed{37.5} \end{aligned}$$

→ Now, select the Node (Line Node) with Maximum Value of Upper Bound.

★ State Space Tree:



→ After step 8, We get one feasible solution

$$\langle 1, 1, 1, 0 \rangle$$

$$\text{Profit} = 32$$

$$\therefore \text{Answer} = \text{Max}(-\infty, 32)$$

$$\therefore \boxed{\text{Answer} = 32}$$

$\therefore$ Answer is updated to 32

→ Kill those Live Nodes whose $\text{LB} < \text{Answer}$ i.e. 32

At present, None of the live Nodes will be Killed.

→ Now, Node 7 will become E-Node.

→ At Node 9, $\text{UB} = 38$ & It is Leaf

$$\therefore \text{Answer} = \text{Max}(\text{Answer}, \text{UB of the New Sol}^n)$$

$$= \text{Max}(32, 38)$$

$$= \boxed{38} \quad \text{Sol}^n = \underline{\underline{\langle 1, 1, 0, 1 \rangle}}$$

→ Kill those Live Nodes

whose $\text{UB} < 38$ ($\because 38$ is the Best Solution so far)

∴ Kill Node 3, 5

$$(\because 37.5 < 38, 36 < 38)$$

→ Last Create Node 10.

$$UB = 20 + (15 - 6) \times 6$$
$$= 20$$

Again it will be killed because
 $20 < 38$.

→ Now, List of Live Nodes is ϕ .

→ ∴ Terminate

∴ Optimal solⁿ = $\langle 1, 1, 0, 1 \rangle$

$$\text{Profit} = \boxed{38}$$